Factors Associated with Systolic Blood Pressure Among Middle-Aged and Older Chinese Adults

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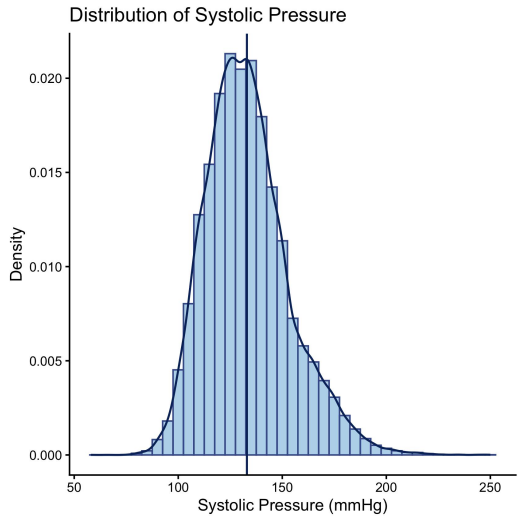
Introduction

Systolic blood pressure (SBP) remains one of the most critical indicators in cardiovascular epidemiology, reflecting the cumulative influence of demographic, behavioral, and metabolic factors. Using a nationally representative survey conducted by Beijing Anzhen Hospital between 2014 and 2016, this study aims to (1) describe the characteristics of the sample population, (2) fit linear models explaining variation in SBP, and (3) evaluate model adequacy and predictive performance. All analyses strictly follow the expectations outlined in the project guidelines, which require a complete *Results* section and a comprehensive *Linear Models* section integrating model formulation, diagnostics, hypothesis testing, and model comparison.

Results

Sample Description

This dataset is derived from a stratified and complex survey conducted among adults aged 45 and above in seven regions of China. A total of 43,105 questionnaires were collected, with a response rate of 90.1%. After excluding 2,278 responses that contained missing values or logical inconsistencies (accounting for 5.28%), the final analysis sample consisted of 40,781 individuals, providing sufficient statistical power for linear modeling. The histogram of blood pressure (SBP) on this page shows a unimodal, approximately normal distribution, slightly skewed to the right, indicating that blood pressure is suitable for modeling under the linear assumption. Further analysis through total sample size correlation calculations further demonstrates that our sample size far exceeds the minimum sample size required to detect a medium effect size, thereby enhancing the reliability of subsequent inferences.



$$f^2 = \frac{R^2}{1 - R^2}$$

$$f^2 = \frac{0.134}{1 - 0.134} = \frac{0.134}{0.866} \approx 0.155$$

$$n = \frac{(Z_{1-\alpha/2} + Z_{1-\beta})^2}{f^2} + k + 1$$

$$Z_{1-\alpha/2} + Z_{1-\beta} = 1.96 + 0.84 = 2.80$$

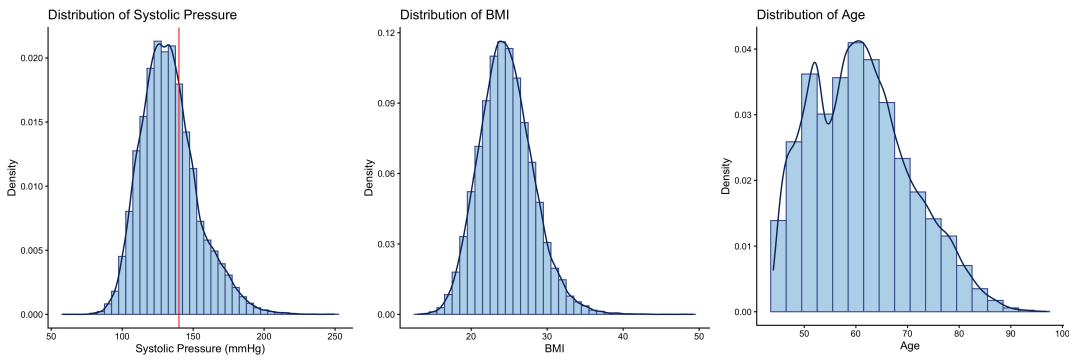
$$(2.80)^2 = 7.84$$

$$n = \frac{7.84}{0.155} + 12 + 1$$

$$n = 50.58 + 13 \approx 63.6$$

Quantitative Variable Distributions

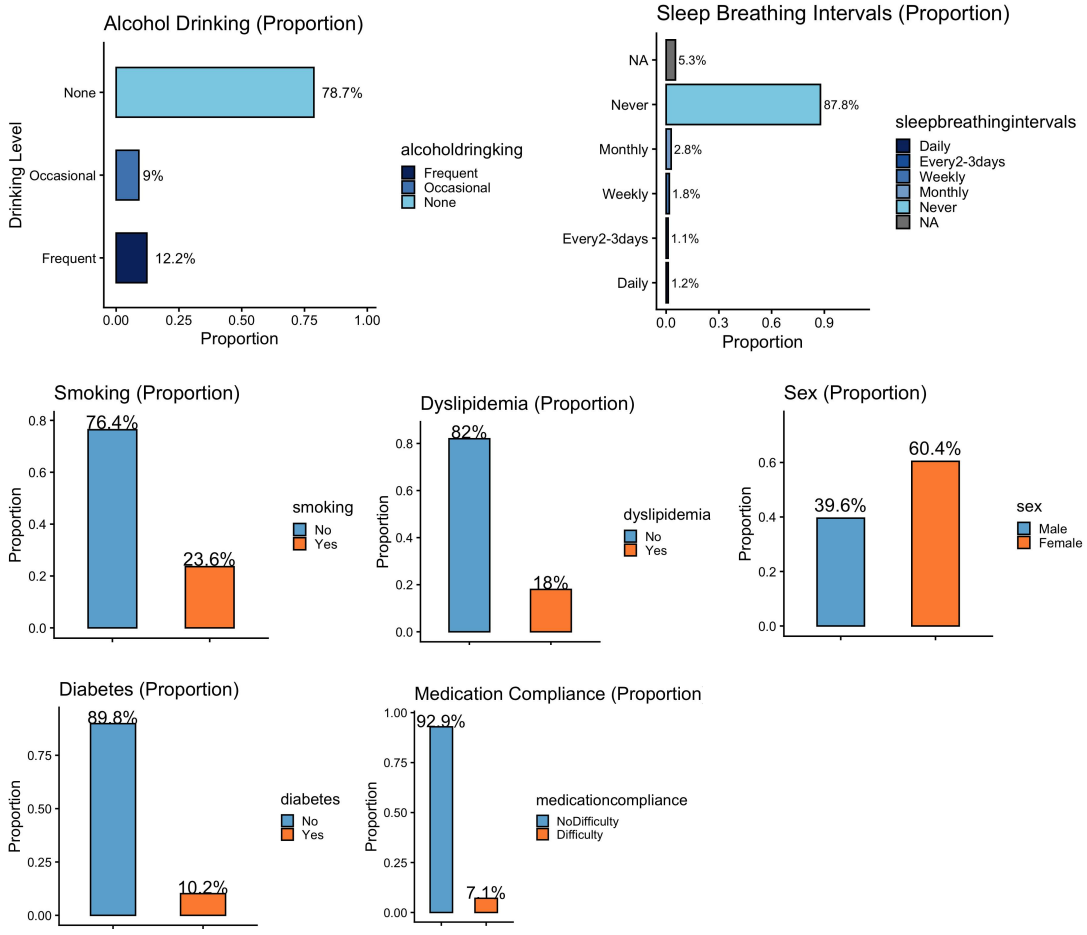
The quantitative variables potentially included in the model are systolic blood pressure (SBP), age, and body mass index (BMI). Systolic blood pressure (as the dependent variable) is concentrated within the clinically significant threshold range emphasized in the "2023 European Society of Cardiology Guidelines", where systolic blood pressure ≥ 140 mmHg is defined as hypertension. Age and body mass index also exhibit a smooth and nearly symmetrical distribution. These visual patterns indicate that there are no extreme outliers or pathological deviations in these quantitative predictive indicators, which supports the hypothesis that these variables make a linear and meaningful contribution to explaining the variation in systolic blood pressure.



Categorical Variable Distributions

We included seven binary and ordinal classification predictors, including gender, smoking status, diabetes condition, dyslipidemia status, alcohol consumption frequency, sleep apnea interval, and medication adherence. There were significant imbalances in some categories - for instance, the proportion of people reporting "no" alcohol consumption was very high - suggesting that there might be some

population-based behavioral patterns that could have different effects on systolic blood pressure in different groups. These charts provided important background information for the encoding selection in the subsequent linear model, particularly regarding the interpretation of reference levels and coefficients.

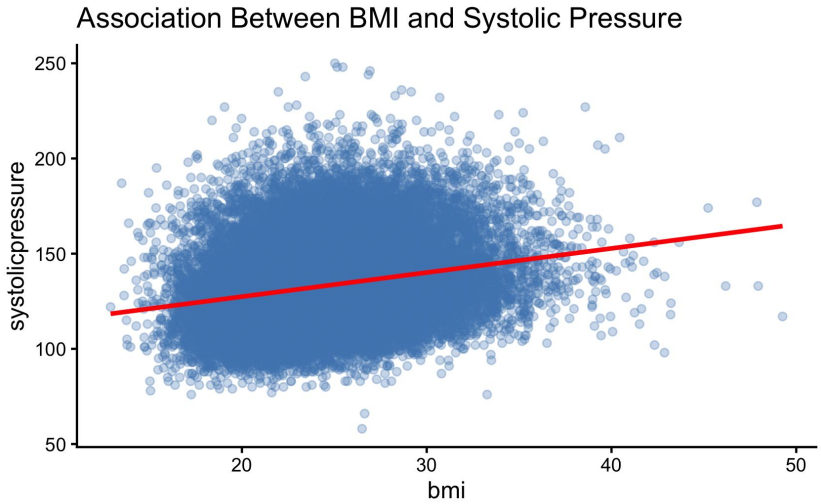


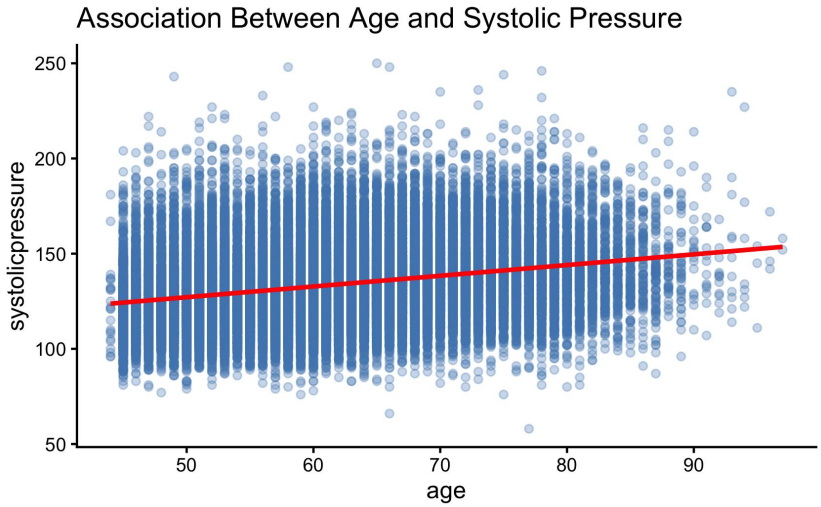
Correlation and Screening Tests

In accordance with the assignment's instruction to “explore summary statistics” and “check for collinearity,” this page reports correlation coefficients and hypothesis tests for all candidate predictors. Age and BMI exhibit positive, statistically significant correlations with SBP ($r = 0.268$ and $r = 0.225$, respectively; both $p < 2e\text{-}16$). Binary predictors such as sex, diabetes, dyslipidemia, and medication compliance demonstrate strong mean differences in SBP based on two-sample t-tests, while smoking shows no significant association ($p = 0.625$). Sleep-breathing intervals also show significant association through ANOVA. These screening results help identify variables likely to contribute meaningfully to the multivariable model.Linear Models

Predictor	Type	Statistical Test	Result (p-value & Direction)
Age	Continuous	Pearson correlation	$r = 0.268$, $p < 2e-16$
BMI	Continuous	Pearson correlation	$r = 0.225$, $p < 2e-16$

Sex	Binary	Two-sample t-test	$p < 2e-16$
Smoking	Binary	Two-sample t-test	$p = 0.625$
Diabetes	Binary	Two-sample t-test	$p < 2e-16$
Dyslipidemia	Binary	Two-sample t-test	$p \approx 3.9e-13$
Medication compliance	Binary	Two-sample t-test	$p < 2e-16$
Alcohol drinking	Ordinal	ANOVA + Linear trend	ANOVA $p = 3.55e-1$; trend $p = 0.891$
Sleep-breathing intervals	Ordinal	ANOVA + Linear trend	ANOVA $p = 0.004$; trend $p = 0.039$





Model Construction

Three nested linear models were constructed:

Model 1: Demographic predictors (sex, age, BMI)

$$\text{systolicpressure}_i = \beta_0 + \beta_1 \boxed{\text{sex}_i} + \beta_2 \boxed{\text{age}_i} + \beta_3 \boxed{\text{bmi}_i} + \epsilon_i$$

Model 2: Adds medical history and behavioral predictors

$$\text{systolicpressure}_i = \beta_0 + \beta_1 \text{sex}_i + \beta_2 \text{age}_i + \beta_3 \text{bmi}_i + \beta_4 \boxed{\text{smoking}_i} + \beta_5 \boxed{\text{diabetes}_i} + \beta_6 \boxed{\text{dyslipidemia}_i} + \beta_7 \boxed{\text{alcoholdringking}_i} + \beta_8 \boxed{\text{sleepbreathingintervals}_i} + \epsilon_i$$

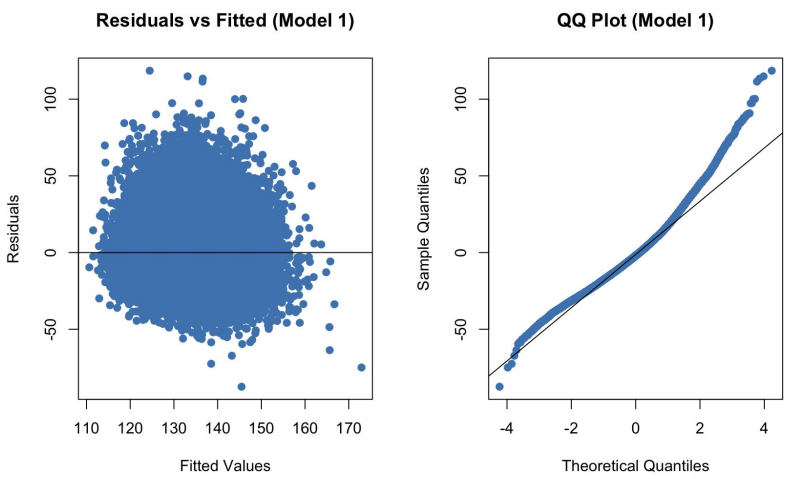
Model 3: Adds clinically motivated interaction terms

$$\begin{aligned}\text{systolicpressure}_i = {} & \beta_0 + \beta_1 \text{sex}_i + \beta_2 \text{age}_i + \beta_3 \text{bmi}_i + \beta_4 \text{smoking}_i + \beta_5 \text{diabetes}_i + \beta_6 \text{dyslipidemia}_i + \beta_7 \text{alcoholdringking}_i + \beta_8 \text{sleepbreathingintervals}_i + \beta_9 \text{medicationcompliance}_i \\ & + \beta_{10} \boxed{(\text{diabetes}_i \times \text{medicationcompliance}_i)} + \beta_{11} \boxed{(\text{smoking}_i \times \text{dyslipidemia}_i)} + \beta_{12} \boxed{(\text{sleepbreathingintervals}_i \times \text{bmi}_i)} + \epsilon_i\end{aligned}$$

This hierarchical structure (M1 ⊂ M2 ⊂ M3) allows for formal model comparison via F-tests, as expected in the assignment. The full model in Page 6 visually outlines each added term, reinforcing the rationale for including interaction effects such as diabetes × medication compliance and smoking × dyslipidemia, both of which carry biological plausibility in cardiovascular physiology.

Model 1: Estimates and Diagnostics

Model 1 demonstrates that sex, age, and BMI are all significant predictors of SBP. The residual-versus-fitted plot exhibits no discernible curvature or heteroskedasticity, while the QQ plot shows only mild right-tail deviation, which is acceptable given the large sample size. Variance inflation factors (all near 1) confirm minimal collinearity. Together, these results satisfy the guideline requirement to “discuss the extent to which the model’s assumptions are met.”




```
Call:
lm(formula = systolicpressure ~ sex + age + bmi, data = df)
Residuals:
   Min     1Q Median     3Q    Max
-87.47 -12.95  -1.99  10.46 118.54

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 65.954208   0.888884  74.199  < 2e-16 ***
sexFemale   -0.796986   0.186468  -4.274 1.92e-05 ***
age          0.576962   0.009506  60.692  < 2e-16 ***
bmi          1.324252   0.025400  52.136  < 2e-16 ***
Residual standard error: 18.82 on 43055 degrees of freedom
Multiple R-squared:  0.1274,    Adjusted R-squared:  0.1274
F-statistic:  2096 on 3 and 43055 DF,  p-value: < 2.2e-16
Anova Table (Type II tests)

Response: systolicpressure
            Sum Sq    Df  F value    Pr(>F)
sex           6469     1   18.268 1.923e-05 ***
age        1304353     1 3683.510 < 2.2e-16 ***
bmi         962526     1 2718.184 < 2.2e-16 ***
Residuals 15246037 43055
```

Model 1: Global F-Test

We conducted a complete hypothesis test for overall model significance in this level:

H_0 : $\beta 1 = \beta 2 = \beta 3 = 0$

H_1 : At least one coefficient is non-zero

Using calculated SSE, SSR, RSE, and R^2 values ($R^2 = 0.1274$), the resulting F-statistic (≈2096) far exceeds the critical value (≈2.60). With $p < 2.2e\text{-}16$, the null hypothesis is rejected. This presentation mirrors the formal test-writing structure illustrated in the sample paper.

```
Call:
lm(formula = systolicpressure ~ sex + age + bmi, data = df)
```

1. Hypothesis

H0: $\beta_{sex} = \beta_{age} = \beta_{bmi} = 0$

H1: at least 1 $\beta_i \neq 0$

2.Test Statistics

$$F = \frac{(\frac{SSR}{p})}{(\frac{SSE}{n-p-1})}$$

$$F = 2096$$

3.Critical Value

$$F_{0.95}(df_1 = 3, df_2 = 43055) \approx 2.60$$

4.Conclusion

Since $F_{model} >> F_{criticalvalue}$

We could reject Null hypothesis

Which could also be confirmed by p-value<2.2e-16

1.SSE:

$$RSE = \sqrt{\frac{SSE}{df}}$$

$$SSE = RSE^2 \times df$$

$$SSE = 15246037$$

2.SSR:

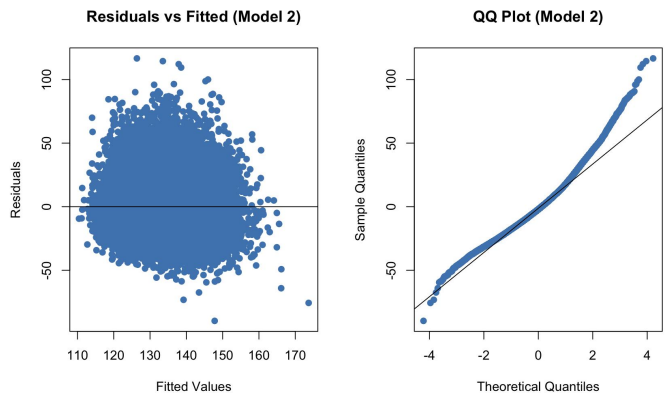
$$R^2 = \frac{SSR}{SST}$$

$$SST = \frac{SSE}{1 - R^2}$$

$$SSR = R^2 \times SST = 227335348$$

Model 2: Expanded Predictors

Model 2 introduces medical conditions and behavioral factors, many of which are strongly associated with SBP (e.g., diabetes, dyslipidemia, medication compliance). Diagnostics remain acceptable, and VIF values indicate only slight increases in collinearity. The addition of these variables improves model explanation, illustrating the value of incorporating clinically meaningful predictors.




```
Call:
lm(formula = systolicpressure ~ sex + age + bmi + smoking + diabetes +
    dyslipidemia + alcoholdringking + sleepbreathingintervals,
    data = df)
Residuals:
    Min      1Q  Median      3Q     Max
-89.799 -12.959  -2.062  10.494 116.626

Coefficients:
                          Estimate Std. Error t value Pr(>|t|)
(Intercept)               65.59883    0.99772  65.749  < 2e-16 *
sexFemale                 -0.54658    0.23149  -2.361  0.01822 *
age                        0.57999    0.00999  58.059  < 2e-16 *
bmi                        1.34597    0.02663  50.540  < 2e-16 *
smokingYes                 0.49107    0.26430   1.858  0.06318 .
diabetesYes                1.94831    0.31759   6.135 8.61e-10 *
dyslipidemiaYes           -1.34315    0.24999  -5.373 7.80e-08 *
alcoholdringking.L        -0.69400    0.21617  -3.211  0.00133 *
alcoholdringking.Q         1.58255    0.28080   5.636 1.75e-08 *
sleepbreathingintervals.L -0.28574    0.62265  -0.459  0.64630
sleepbreathingintervals.Q -0.24450    0.63895  -0.383  0.70198
sleepbreathingintervals.C -1.86598    0.69619  -2.680  0.00736 *
sleepbreathingintervals^4  1.50588    0.69498   2.167  0.03026 *
Residual standard error: 18.87 on 40768 degrees of freedom
  (2278 observations deleted due to missingness)
Multiple R-squared:  0.1311,    Adjusted R-squared:  0.1308
F-statistic: 512.4 on 12 and 40768 DF,  p-value: < 2.2e-16
```

Model 3: Interaction Terms

We evaluates three interaction effects in model3:

> Diabetes × Medication Compliance: The strong negative interaction ($\beta \approx -4.46$) indicates that poor medication compliance elevates SBP particularly among individuals with diabetes.
>
> Smoking × Dyslipidemia: The positive interaction ($\beta \approx 1.97$) aligns with the compounding vascular burden seen clinically.
>
> Sleep-Breathing Intervals × BMI: A weaker but conceptually consistent interaction.

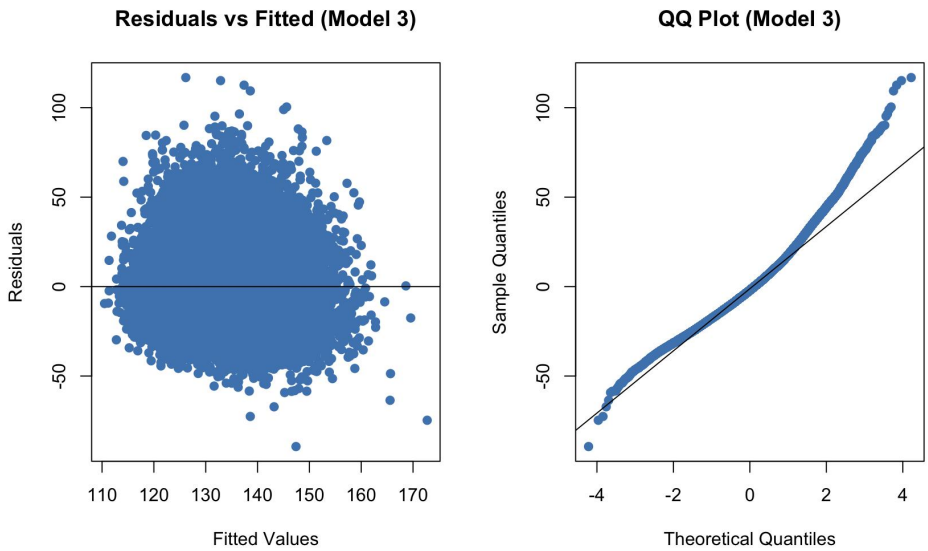
These results demonstrate thoughtful integration of clinical mechanism into statistical modeling, paralleling the example paper's analytic rationale.

```
Residuals:
    Min      1Q  Median      3Q     Max 
-89.419 -12.905  -2.042  10.544 116.844 

Coefficients:
                                          Estimate Std. Error t value Pr(>|t|)    
(Intercept)                             68.6220027  2.1997497  31.195  < 2e-16 ***
sexFemale                               -0.6176543  0.2312320  -2.671 0.007562 ** 
age                                      0.5720186  0.0100063  57.166  < 2e-16 ***
bmi                                      1.2384533  0.0832797  14.871  < 2e-16 ***
smokingYes                               0.1024542  0.2796563   0.366 0.714100    
diabetesYes                              2.2332645  0.3463342   6.448 1.14e-10 ***
dyslipidemiaYes                         -1.9931471  0.2779496  -7.171 7.58e-13 ***
alcoholdringking.L                      -0.7377046  0.2158236  -3.418 0.000631 ***
alcoholdringking.Q                       1.6285558  0.2804280   5.807 6.39e-09 ***
sleepbreathingintervals.L               -8.3607904  4.4908217  -1.862 0.062645 .  
sleepbreathingintervals.Q                3.1532441  4.5772405   0.689 0.490892    
sleepbreathingintervals.C                1.7299227  4.7707048   0.363 0.716895    
sleepbreathingintervals^4               -0.0004023  4.8094410   0.000 0.999933    
medicationcomplianceDifficulty           5.0596150  0.4207559  12.025  < 2e-16 ***
diabetesYes:medicationcomplianceDifficulty -4.4645080  0.8721128  -5.119 3.08e-07 ***
smokingYes:dyslipidemiaYes               1.9735823  0.6047995   3.263 0.001103 ** 
bmi:sleepbreathingintervals.L            0.3214298  0.1785056   1.801 0.071762 .  
bmi:sleepbreathingintervals.Q           -0.1324978  0.1827558  -0.725 0.468457    
bmi:sleepbreathingintervals.C           -0.1417405  0.1895055  -0.748 0.454495    
bmi:sleepbreathingintervals^4            0.0588466  0.1924879   0.306 0.759823    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 18.84 on 40761 degrees of freedom
  (2278 observations deleted due to missingness)
Multiple R-squared:  0.1345,	Adjusted R-squared:  0.1341 
F-statistic: 333.3 on 19 and 40761 DF,  p-value: < 2.2e-16
```

Model 3: ANOVA and Diagnostics

The Type-II ANOVA table highlights which predictors retain significance after adjusting for others. Residual plots remain acceptable, suggesting that the increased complexity of Model 3 does not compromise model assumptions. The highlighted F-values underscore the importance of several interaction effects.

```
Anova Table (Type II tests)

Response: systolicpressure
                                Sum Sq    Df   F value    Pr(>F)
sex                               2531     1    7.1350  0.007562
age                            1159380     1 3267.9630 < 2.2e-16
bmi                             886043     1 2497.5045 < 2.2e-16
smoking                            832     1    2.3439  0.125779
diabetes                          8304     1   23.4056 1.317e-06
dyslipidemia                     14476     1   40.8024 1.702e-10
alcoholdringking                 13375     2   18.8508 6.562e-09
sleepbreathingintervals           5465     4    3.8512  0.003936
medicationcompliance             42208     1  118.9731 < 2.2e-16
diabetes:medicationcompliance     9297     1   26.2061 3.082e-07
smoking:dyslipidemia              3778     1   10.6485  0.001103
bmi:sleepbreathingintervals       1286     4    0.9064  0.459043
Residuals                     14460840 40761
```



Model Comparison

Nested ANOVA Comparison

Formal comparison of nested models shows:

> Model 1 vs Model 2: Highly significant improvement ($p < 2.2e-16$).
>
> Model 2 vs Model 3: Additional interactions also significantly improve model performance.

This directly fulfills the assignment's requirement to "compare your model to either a larger or smaller model" using an appropriate F-test.

```
Model 1: systolicpressure ~ sex + age + bmi
Model 2: systolicpressure ~ sex + age + bmi + smoking + diabetes + dyslipidemia +
    alcoholdringking + sleepbreathingintervals
  Res.Df      RSS Df Sum of Sq      F    Pr(>F)
1  40777 14558229
2  40768 14517798  9     40432 12.615 < 2.2e-16 ***
```

```
Model 1: systolicpressure ~ sex + age + bmi + smoking + diabetes + dyslipidemia +
    alcoholdringking + sleepbreathingintervals
Model 2: systolicpressure ~ sex + age + bmi + smoking + diabetes + dyslipidemia +
    alcoholdringking + sleepbreathingintervals + medicationcompliance +
    diabetes:medicationcompliance + smoking:dyslipidemia + sleepbreathingintervals:bmi
  Res.Df      RSS Df Sum of Sq      F    Pr(>F)
1  40768 14517798
2  40761 14460840  7     56958 22.935 < 2.2e-16 ***
```

Information Criteria and Fit Statistics

Adjusted R^2 increases monotonically from Model 1 to Model 3 (0.1286 → 0.1341).
AIC and BIC both decrease across models, indicating improved model fit even after penalizing complexity.
Residual standard error decreases as well.

These patterns collectively support Model 3 as the best-performing model.

Model	Adj R^2	AIC	BIC	Residual SE
Model 1	0.1286	355439.8	355482.8	18.895
Model 2	0.1308	355344.3	355465.0	18.871
Model 3	0.1341	355198.0	355379.0	18.835

Detailed F-Test: Model 2 vs Model 3

We conducted a complete hypothesis test for model compare:

$\Delta RSS = 56{,}958$

$\Delta df = 7$

MSE for Model 3 = 35,494

These yield an F-statistic ≈22.94, far above the critical value of 2.01. Thus, the additional interaction terms significantly improve model explanatory power.

1. Hypothesis

H0: All $\beta_i = 0$

H1: at least 1 $\beta_i \neq 0$

2.Test Statistics

$$F = \frac{\left(\frac{RSS_2 - RSS_3}{df_2 - df_3}\right)}{\frac{RSS_3}{df_3}}$$

$$F \approx 22.935$$

3.Critical Value

$$F_{0.95}(7,40161) \approx 2.01$$

4.Conclusion

Since $F_{model} > F_{criticalvalue}$

We could reject Null hypothesis

We have:

$RSS_2 = 14517798$

$RSS_3 = 14460840$

$df_2 = 40768$

$df_3 = 40761$

1.Differences in RSS:

$$RSS_2 - RSS_3 = 14517798 - 14460840 = 56958$$

2.Differences in df:

$$df_2 - df_3 = 40768 - 40761 = 7$$

3.MSE:

$$\frac{RSS_3}{df_3} = 35494$$

Confidence Intervals for New Observations

Prediction intervals for new observations demonstrate an important distinction between model fit and predictive precision. Model 1 produces the narrowest interval (width ≈0.71), whereas Model 3, despite best overall fit, exhibits the widest interval (≈1.38). This widening reflects increased variance introduced by added predictors and interactions—especially relevant in large multivariable models where coefficient instability may propagate into predictions.

Model	Mean Point	CI Lower	CI Upper	CI Width
Model 1	133.30	132.94	133.65	0.71
Model 2	133.30	132.70	133.90	1.21
Model 3	133.25	132.55	133.94	1.38

Conclusion

Through structured exploratory analysis and sequential model development, this study identifies demographic, behavioral, and metabolic factors associated with systolic blood pressure among older Chinese adults. The hierarchical modeling strategy reveals that interaction effects—particularly those involving diabetes, medication compliance, smoking, and dyslipidemia—substantially enhance explanatory power. However, the increased complexity also reduces predictive precision, underscoring the tradeoff between model fit and variance inflation. Overall, Model 3 provides the most comprehensive description of SBP variation in this population, aligning with both statistical best practices and physiological understanding.